Certificate of Attendance

This is to certify that Mr.

David Pelosi

has completed the

"Second CNCG School on Global Navigation Satellite Systems and Related Technologies"

The course, which lasted a total of 42 hours, was conducted in a hybrid format, with both remote and in-person sessions held at Telespazio's headquarters on April 9, 10, 16, 17, and 21, 2026, for the online sessions, and on April 23 and 24, 2026, for the on-site sessions.

TELESPAZIO S.p.A.

Francesco Malvolti
(PM CNCG PROGRAMME)

Rome, 05/05/2026

<table>
<tr><td colspan="2">ACHIEVED COMPETENCES
Understanding of the fundamental principles of satellite positioning, the structure of GNSS constellations, the architecture and signals of receivers, and performance metrics along with correction and augmentation techniques.</td></tr>
<tr><td colspan="2">LEARNING OUTCOMES</td></tr>
<tr><td><u>KNOWLEDGE</u>
• Fundamentals of GNSS radionavigation and receiver state estimation techniques.
• GNSS performance metrics, reference systems, and orbital mechanics of major constellations.
• Signal structures, navigation message formats, and multi-constellation signal characteristics.
• Error sources analysis including clock, atmospheric, multipath, and transmission delays.
• Augmentation systems (SBAS, EGNOS, RTK, PPP-RTK) and integrity monitoring concepts.
• Vulnerabilities of GNSS signals including jamming and spoofing attack mechanisms</td><td><u>SKILLS</u>
• Ability to perform GNSS error analysis including clock, atmospheric, multipath, and transmission delays.
• Skills in performance evaluation using standard GNSS metrics and reference systems.
• Competence in interpreting GNSS signal structures and navigation message formats.
• Expertise in error budget assessment for GNSS links (power, fading, interference).
• Improved capability in configuring and analyzing GNSS corrections (clock, code, atmospheric).
• Enhanced proficiency in vulnerability assessment of GNSS signals against jamming and spoofing.
• Collection and analysis of GNSS raw observables using Android GNSS Logger and Google post-processing tools
• Implementation of Least-Squares multilateration algorithms with Gauss-Newton iterative solver.
• Practical operation of professional GNSS laboratory equipment and signal generators.</td></tr>
<tr><td colspan="2">LEARNING VERIFICATION METHODS</td></tr>
<tr><td colspan="2">CLASSROOM: 42 hours</td></tr>
<tr><td>EVALUATION METHODS
X Written Test
Oral Test
Practical Test
Performance Test</td><td>COMPETENCE LEVEL
Basic
X Mature
Expert</td></tr>
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